

VHF-Sim2Real: Generative Noise Injection for Improved Automatic Speech Recognition in Air Traffic Control

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Robust Automatic Speech Recognition (ASR) is a prerequisite for automated support tools in Air Traffic Control (ATC), yet publicly available labeled corpora of real Very High Frequency (VHF) radio traffic are extremely limited. The open portion of the ATCO2 corpus provides only about one hour of manually transcribed VHF speech, whereas the widely used ATCOSIM corpus contains roughly ten hours of clean, simulated controller speech recorded in a laboratory. After fine-tuning on the combination of ATCOSIM and the labeled ATCO2 subset, large open ASR models such as Whisper can reach a Word Error Rate (WER) of about 1.2% on ATCOSIM, yet WER on real ATCO2 speech remains around 13–15% as reported by van Doorn et al. and confirmed by our reproduction. This behavior highlights both the acoustic mismatch between simulated and real VHF channels and the limited amount of labeled operational data. We investigate whether generative noise modeling can reduce this mismatch without introducing additional labeled ATC speech. Our *VHF-Sim2Real* pipeline trains a spectrogram-based Generative Adversarial Network (GAN) on unlabeled VHF recordings from TartanAviation and ATCO2 to learn VHF-like interference patterns and spectral distortions, then uses the trained generator to inject this noise on the fly into ATCOSIM utterances during Whisper fine-tuning. The fine-tuning stage mixes these noisy ATCOSIM signals with about one hour of labeled ATCO2 speech, so that Whisper is exposed to ATC phraseology under realistic VHF degradations while still benefiting from accurate transcripts. On the evaluation split of the public ATCO2-test-set-1h subset, this procedure reduces WER from 14.32% for our reproduction of the Whisper-ATC baseline to 3.58%, a 75.0% relative improvement obtained with the same amount of labeled ATC data, and suggests that, in this benchmark setting, unpaired generative noise augmentation can substantially reduce the sim-to-real performance gap. Code and repository are available at <https://github.com/omargarib02/VHF-Sim2Real>.

I. Introduction

CLEAR voice communication between air traffic controllers (ATCs) and pilots is essential for aviation safety. Rigorous communication protocols have been established and continuously refined to meet evolving safety requirements and reduce operational risk [1]. However, continued growth in air traffic volumes and increasingly complex airspace management, for example at airports that serve both commercial airlines and military operations, make effective communication more challenging [2, 3]. At major high-traffic airports, controllers frequently handle multiple clearances simultaneously and rely heavily on memory to track rapidly changing situations [4]. Airports with shared civil and military airspace add further complexity through diverse aircraft types and mission profiles, increasing cognitive load and tightening the margins for safe, unambiguous communication [3]. Although controllers are extensively trained and generally equipped to manage such demanding situations, errors can still occur under high workload and cognitive strain. Even minor miscommunications, such as a missed altitude clearance or an incorrectly acknowledged call sign, can escalate into serious incidents including runway incursions or loss-of-separation events [4, 5].

Providing controllers with additional support or oversight mechanisms could help reduce the risk of such miscommunications. One natural direction is an AI-based assistant that continuously monitors communications, verifies pilot read-backs, and flags potential discrepancies [6]. Developing such assistants first requires accurate transcription of controller–pilot exchanges. Real-time Automatic Speech Recognition (ASR) is the core technology for capturing these spoken interactions, but existing ASR models often struggle under the noisy Very High Frequency

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(VHF) radio conditions typical of operational ATC environments [7]. Modern large-vocabulary ASR models such as OpenAI’s Whisper [8] perform well on a broad range of benchmark datasets, yet their accuracy drops sharply when evaluated on realistic ATC audio [9].

A central obstacle to developing robust ATC ASR is the limited availability of labeled real-world radio data [10]. For example, the ATCO2 project publicly releases about one hour of manually transcribed test data from real VHF communications, together with a much larger but weakly labeled pool of several thousand hours [10]. In contrast, simulation corpora such as ATCOSIM [11] provide about ten hours of clean, scripted ATC speech recorded with professional controllers in laboratory conditions. These simulated utterances are easier to script and transcribe than live VHF traffic, yet they do not exhibit the full range of artifacts encountered on operational channels [7, 10].

Recent work by van Doorn et al. quantifies both the benefits and the limitations of combining these datasets for Whisper fine-tuning [9]. Starting from an out-of-the-box word error rate (WER) of 24.03% on ATCO2 and 16.74% on ATCOSIM, they fine-tune Whisper large-v2 on three settings: only the 0.86-hour labeled ATCO2 training partition, only the 8.37 hours of ATCOSIM, and the combined 9.23 hours. Fine-tuning on ATCO2 alone reduces WER on ATCO2 to 14.66%, whereas adding ATCOSIM yields only a modest further improvement to 13.46% on ATCO2 while achieving 1.17% WER on ATCOSIM (compared with 1.19% when training on ATCOSIM alone). In our reproduction on the public ATCO2 test set, the corresponding Whisper-ATC configuration yields a WER of 14.32%. Together, the results of van Doorn et al. and our own reproduction indicate that adding roughly ten hours of clean simulated speech to less than one hour of labeled VHF audio yields only modest gains, which points to the remaining acoustic mismatch between laboratory speech and real VHF communications as the primary limitation.

In this work, we investigate whether generative modeling of VHF noise can reduce this mismatch without requiring additional manual transcription. We introduce *VHF-Sim2Real*, a noise-augmentation pipeline that learns the spectral characteristics of VHF radio channels from unlabeled recordings and uses them to perturb simulated ATC speech during fine-tuning. We train a Generative Adversarial Network (GAN) on unpaired magnitude spectrograms from large unlabeled corpora of ATC radio, including TartanAviation [12], which provides several thousand hours of terminal-area ATC speech, and the raw ATCO2 pool. The generator learns to map clean spectra to spectra that carry the texture of real VHF interference. During Whisper fine-tuning, we apply this generator on the fly only to ATCOSIM utterances, while using the transcribed ATCO2 training set as additional clean supervision. This yields additional noisy variants of each labeled simulation utterance, so that the model is exposed to ATC phraseology under channel conditions that more closely resemble real operations.

Our primary contributions are:

- 1) **Unpaired Noise Modeling.** We develop a GAN-based pipeline that learns VHF noise characteristics such as squelch and static from *unlabeled* real-world ATC audio, avoiding the need for parallel clean–noisy recordings.
- 2) **Generative Augmentation.** We use the trained generator to inject realistic VHF-like noise into ATCOSIM utterances during training, increasing the acoustic diversity of a limited labeled simulation corpus while preserving the original transcripts.
- 3) **ASR Performance Gains.** We show that fine-tuning Whisper on this augmented mixture of ATCOSIM and ATCO2 reduces WER from 14.32% for our reproduction of the Whisper-ATC baseline of van Doorn et al. to 3.58% on the ATCO2 test set, indicating that generative noise augmentation can substantially narrow the gap between simulated and real ATC speech.

The remainder of this paper is organized as follows. Section II reviews the acoustic properties of ATC communication, relevant ASR architectures, and prior work on speech enhancement and data augmentation. Section III describes the corpora used in this study. Section IV introduces the VHF-Sim2Real pipeline, and Section V details the training and evaluation setup. Section VI presents quantitative and qualitative evaluation on the ATCO2 benchmark. Section VII discusses the main limitations of our study, and Section VIII outlines directions for extending this work. Section IX concludes the paper.

II. Background

Air Traffic Control (ATC) relies heavily on voice communication to maintain airspace safety and efficiency. However, operational environments introduce unique challenges, including spectral congestion, non-standard phraseology, and high-energy background noise, that degrade the performance of general-purpose speech recognition systems. This section reviews the specific acoustic characteristics of ATC communication, the evolution of ASR architectures, and the role of generative modeling in addressing data scarcity.

A. Real-Time Challenges in ATC Voice Communications

ATC voice communication takes place in an operational setting where controllers manage multiple aircraft at once and safety margins depend on timely, unambiguous exchanges. At busy terminals, controllers work under high cognitive loads and must coordinate traffic flows that can include mixed civil and military operations, which increases procedural complexity [3]. While standardized phraseology is mandated to minimize ambiguity, operational analyses and safety briefings indicate that high workload, fatigue, and frequency congestion are associated with incomplete communication, nonstandard phraseology, and missed readback corrections [4, 5]. In addition, accent diversity, variation in speaking rate, and non-standard abbreviations have been shown to contribute to misunderstanding between pilots and controllers [13, 14].

From an acoustic perspective, the radio channel adds its own difficulties. VHF transmissions are band limited to approximately 300–3400 Hz and are frequently affected by non-stationary interference [15]. In practice, controllers and pilots often contend with blocked or “stepped-on” transmissions when two aircraft key the frequency at the same time, brief squelch bursts as the carrier opens or closes, and clipped phrases when microphones are keyed too late or released too early [1, 16]. Cockpit noise from engines and airflow further reduces the effective signal-to-noise ratio of pilot transmissions [1].

Collectively, these linguistic and acoustic effects define a domain that differs systematically from the curated corpora and benchmark conditions under which many general-purpose ASR models are developed and evaluated. Even within ATC research, widely used simulated corpora such as ATCOSIM are much easier to produce and transcribe than operational recordings, but they do not capture the full range of artifacts present on real VHF channels [7, 10, 11]. Van Doorn et al. show that fine-tuning Whisper on roughly nine hours of labeled ATC speech, the majority of which is simulated ATCOSIM audio, yields word error rates close to one percent on the simulated domain, while performance on real ATCO2 speech remains in the mid-teens [9]. In parallel, earlier benchmarks on ATC speech show that substantial training data does not eliminate recognition errors on real radio communications. Zuluaga-Gomez et al. train DNN-based systems on roughly 176 hours of in-domain ATC audio together with about 1500 hours of out-of-domain read speech, yet WER on some realistic ATC test sets (e.g., Airbus) remains around 14–15% [7]. For the ATCO2 corpus, baseline systems trained on approximately 5281 hours of ATC recordings still obtain WERs above 20% on the 4-hour ATCO2 test set [10].

B. Evolution of ASR Architectures

Modern ASR systems for domains such as ATC build on several generations of architectural development, moving from hybrid Hidden Markov Model–Gaussian Mixture Model (HMM-GMM) pipelines to large end-to-end neural models. Early deep learning work replaced GMMs with Deep Neural Networks (DNNs) in the acoustic model [17], which led to substantial accuracy gains while keeping the HMM-based decoding framework intact. This hybrid stage was followed by sequence-to-sequence architectures, such as Listen, Attend and Spell (LAS) [18] and Recurrent Neural Network Transducers (RNN-T) [19], which removed the need for explicit frame-level alignments and learned a direct mapping from acoustic feature sequences to output symbols.

Subsequent work shifted toward attention-based models. The Transformer [20] introduced self-attention as a replacement for recurrent connections, allowing models to capture long-range dependencies and global context more effectively, while architectures such as the Conformer [21] combined convolutional modules with self-attention to better represent local and long-range structure in speech. In parallel, self-supervised frameworks like wav2vec 2.0 [22] showed that robust speech representations could be learned from large amounts of unlabeled audio and then adapted to downstream ASR tasks with relatively little labeled data. These trends toward attention-based sequence modeling and large-scale pretraining are reflected in the current generation of general-purpose ASR models.

The Whisper Framework. OpenAI’s Whisper [8] is an open-weight Transformer encoder–decoder model trained on approximately 680,000 hours of weakly supervised multitask audio. The breadth of its training data, which spans diverse languages, speakers, and acoustic conditions, yields strong zero-shot performance in many settings compared with models trained only on curated corpora such as LibriSpeech [23]. In the ATC domain, however, Whisper exhibits substantially higher error rates on speech transmitted over VHF radio channels than on cleaner benchmark data [9]. Van Doorn et al. [9] reported that fine-tuning Whisper on clean simulated ATC speech improves recognition of domain-specific phraseology but reduces robustness to real VHF noise. In this work, we therefore seek an adaptation strategy that specializes Whisper to ATC while maintaining robustness to VHF channel degradations by exposing the model to both ATC-specific language and the spectral characteristics of radio communication during fine-tuning.

C. Deep Learning for Speech Enhancement

Speech enhancement aims to improve the intelligibility and perceived quality of speech corrupted by background noise. Classical single-channel methods rely on spectral subtraction, Wiener filtering, and related statistical estimators [24]. These approaches are effective for approximately stationary noise but tend to degrade in rapidly changing environments. Neural models are now widely used because they can learn non-linear time–frequency masks and spectral mappings from data containing non-stationary interference. Early generative work such as SEGAN [25] introduced a waveform-level GAN that outperformed conventional baselines, including Wiener filtering, on both objective and perceptual quality measures. Real-time systems like Valin’s hybrid digital signal processing (DSP) and deep learning approach (RNNoise) [26] further showed that small recurrent networks combined with classical signal processing can provide full-band noise suppression with low computational cost. Subsequent phase-aware architectures, for example DCCRN [27], use complex-valued convolutional recurrent networks to model magnitude and phase jointly and achieved top performance in the Interspeech 2020 Deep Noise Suppression challenge.

In aviation applications, several studies adapt these ideas to the radio environment. Chen et al. [28] proposed a deep convolutional Wasserstein GAN for flight crew dialog recorded under simulated aircraft operating conditions. Their method improves background-noise intrusiveness and overall perceptual quality compared with a WGAN baseline. Liang et al. [29] introduced DeCGAN, a MetricGAN-style model whose generator is a DeConformer module with time, frequency, and channel attention and deformable convolution feed-forward blocks. On an ATC corpus they obtained competitive PESQ and STOI scores for civil aviation control speech, illustrating that GAN-based enhancement can be tailored directly to controller communications.

Building on these ATC-specific enhancement efforts, ROSE [30] is a time-domain enhancement network for ATC that uses a U-Net architecture with attention-based skip fusion and channel and sequence attention modules. It is trained with a combination of signal-level and ASR-oriented losses to improve both intelligibility and word error rate while keeping the downstream recognizer fixed. In parallel, noise-robust Unified Two-pass (U2) schemes for ATC ASR [31] adapt unified two-pass end-to-end architectures to air traffic voice communication, with a focus on maintaining recognition accuracy under strong radio noise. Our work differs from these studies in how GANs are used: rather than inserting a learned enhancement front-end before the recognizer, we train a generator to synthesize realistic VHF noise for data augmentation and then fine-tune Whisper directly on the resulting mixtures.

D. Evolution of Data Augmentation Strategies

In recent ASR systems, data augmentation has progressed from simple deterministic operations on the waveform to feature-domain and learned approaches. Early work applied speed perturbation [32] and convolution with Room Impulse Responses (RIRs) to simulate different speakers and acoustic environments [33]. Large-scale noise corpora such as MUSAN [34] complemented these techniques by providing diverse background audio for additive mixing, which allowed models to encounter a broader range of signal-to-noise ratios and noise types during training. Building on these ideas, later work extended augmentation to the feature domain. SpecAugment [35] masks contiguous blocks of time and frequency in the spectrogram and yields consistent robustness improvements on several benchmarks. Mixup-style approaches such as MixSpeech [36] extend this feature-domain view by interpolating pairs of utterances and their labels, training the recognizer to predict both transcripts and outperforming SpecAugment on a set of low-resource tasks. More recent studies explore task-specific augmentation, for example using text-to-speech (TTS) systems to synthesize additional training utterances for low-resource ASR [37] and designing acoustic-aware augmentation policies that explicitly target the robustness of large end-to-end models [38].

Although these approaches remain effective in many settings, they show limitations when applied directly to ATC. Heuristic masking such as SpecAugment regularizes the encoder but does not encourage the model to separate speech from particular interference patterns, while additive noise injection based on generic noise corpora struggles to reproduce the non-stationary, signal-dependent artifacts of VHF radio, such as squelch bursts and carrier-wave static. To better capture such effects, some recent work has investigated generative models for augmentation and domain adaptation. Variational autoencoder based methods [39] modify latent nuisance attributes to produce additional training examples that move the data distribution closer to a target domain while preserving the underlying transcripts. Diffusion-based speech restoration models [40] learn to reconstruct clean or less corrupted speech under severe distortions, but their iterative sampling procedure makes them relatively expensive for on-the-fly augmentation.

Within this generative line of research, GANs offer an attractive alternative because they generate samples in a single forward pass. For noise-robust ASR, Hu et al. [41] use GANs to synthesize noisy speech that matches a target acoustic domain and report WER improvements over conventional multi-condition training. Chen et al. [42] propose

SimuGAN, which learns to map clean spectra to noisy spectra using only ten minutes of unpaired in-domain noisy audio and achieves meaningful gains compared with strong baselines. Taken together, these results suggest that generative noise injection can be particularly helpful when only limited in-domain data are available. In our work, we adopt this perspective for aviation by training a GAN on unlabeled VHF recordings to capture the acoustic signature of ATC radio channels and then use the generator to augment clean simulation data during Whisper fine-tuning.

E. Unpaired Domain Translation via GANs

Generative adversarial networks (GANs) [43] are commonly applied to speech enhancement, but most approaches assume parallel noisy-clean pairs so that each noisy utterance has a corresponding clean reference. This assumption does not hold well in the ATC setting. Real VHF recordings are unpaired with respect to simulation data, and there is no clean ground truth for a specific noisy transmission recorded at an airport. As a result, paired image-to-image translation models such as pix2pix [44], which require aligned source and target examples, cannot be applied directly.

We instead treat magnitude spectrograms as images and adopt an unpaired image-to-image translation framework. Cycle-consistent approaches such as CycleGAN [45] address the lack of pairs by learning two generators that map between the clean and noisy domains in opposite directions, together with a cycle-consistency loss that penalizes discrepancies between an original sample and its reconstruction after both mappings. This strategy is effective but increases the number of networks and loss terms that must be optimized. Following SimuGAN [42], which uses a single generator to map clean spectra to noisy spectra under unpaired supervision, we also use a single-direction generator with content-preserving constraints so that linguistic information is retained.

Adversarial training is sensitive to the discriminator parameterization, so prior work commonly applies stabilization techniques such as spectral normalization and PatchGAN-style discriminators [44, 46], strategies that we also adopt in our method.

III. Datasets

Developing robust aviation ASR requires training data that capture both the linguistic structure of controller phraseology and the acoustic conditions of VHF radio. In this work, we rely on three open corpora that span clean simulation, large-scale unlabeled VHF audio, and a small labeled subset of real ATC communications: ATCOSIM [11] provides clean, fully transcribed controller speech from simulations, TartanAviation [12] supplies large-scale unlabeled VHF audio from real operations, and ATCO2 [10] offers a small labeled subset of real ATC communications. Table 1 summarizes the main characteristics and roles of these datasets in our pipeline.

Table 1 Summary of utilized ATC speech corpora. *Role*: (S) Clean Source, (N) Noise Source, (T) Target Domain (Train & Eval). “Total (raw)” denotes the size of the underlying corpus (in number of hours recorded); “Labeled used” denotes the portion with human transcripts used for supervised ASR training.

Dataset	Total (raw)	Labeled used	Role	Acoustic Profile
ATCOSIM [11]	10h	10h	S	Clean simulation
TartanAviation [12]	3,374h	0h	N	Real VHF (high noise)
ATCO2 [10]	5,281h	~1h	N / T	Real VHF (labeled subset)

A. ATCOSIM (Clean Source)

The ATCOSIM corpus [11] serves as our primary source of labeled, clean ATC speech. It contains roughly 10 hours of non-prompted controller utterances recorded during real-time simulations with professional air traffic controllers. Each utterance is transcribed at the word level. Recordings are made under controlled laboratory conditions and are largely free of background noise, radio artifacts, and overlapping speakers, so the audio represents standardized ATC phraseology with high signal quality. We use ATCOSIM as the main source of linguistic content and ground-truth transcripts, which is then combined with synthetic noise during augmentation.

B. TartanAviation (Noise Source)

To model the acoustic conditions of real operations, we use the TartanAviation dataset [12]. This corpus contains approximately 3,374 hours of VHF communications collected around terminal airspace operations and aligned with

trajectory data, but the speech is not manually transcribed at scale. We use TartanAviation exclusively as a source of unlabeled audio when training the GAN. In our listening experiments, the recordings exhibit a wide range of non-stationary channel effects, including variable signal strength, overlapping calls, squelch noise, and strong cockpit and environmental backgrounds, which makes them well suited as a realistic VHF noise reference.

C. ATCO2 (Target Domain)

For supervised fine-tuning and evaluation, we rely on the publicly released labeled subset of the ATCO2 corpus [10]. ATCO2 collects several thousand hours of real ATC voice communications from a network of crowdsourced VHF receivers streaming open radio channels and includes a four-hour ATCO2 test set with carefully verified human transcripts. Of this test set, a one-hour subset (ATCO2-test-set-1h) is publicly available and is the portion we use. This one-hour labeled subset contains exchanges between pilots and controllers, non-native accents, and real VHF channel degradations, and we treat it as our labeled target domain. Following van Doorn et al. [9], we adopt the same 80/20 split of this labeled portion into a training partition and a held-out evaluation partition. The training split is used for supervised fine-tuning in combination with GAN-augmented ATCOSIM and can also contribute real VHF audio to the GAN training data. The evaluation split is kept separate from all training stages, both for Whisper fine-tuning and for GAN training, and is used exclusively to report ASR performance under realistic conditions.

IV. Methodology

We propose *VHF-Sim2Real*, a two-stage generative augmentation pipeline for adapting Whisper to real VHF ATC using limited labeled operational data. The method learns VHF channel noise from unlabeled radio recordings and applies it to clean simulated ATC speech during fine-tuning, effectively expanding the amount of labeled audio that exhibits realistic VHF degradations while preserving the original transcripts. As illustrated in Fig. 1, the pipeline proceeds in two distinct stages: (1) *Stage I: GAN Training*, where a GAN learns to map clean simulation spectrograms to the noisy VHF domain, and (2) *Stage II: Whisper Fine-Tuning*, where Whisper is adapted using a weighted mixture of real and synthetically augmented data.

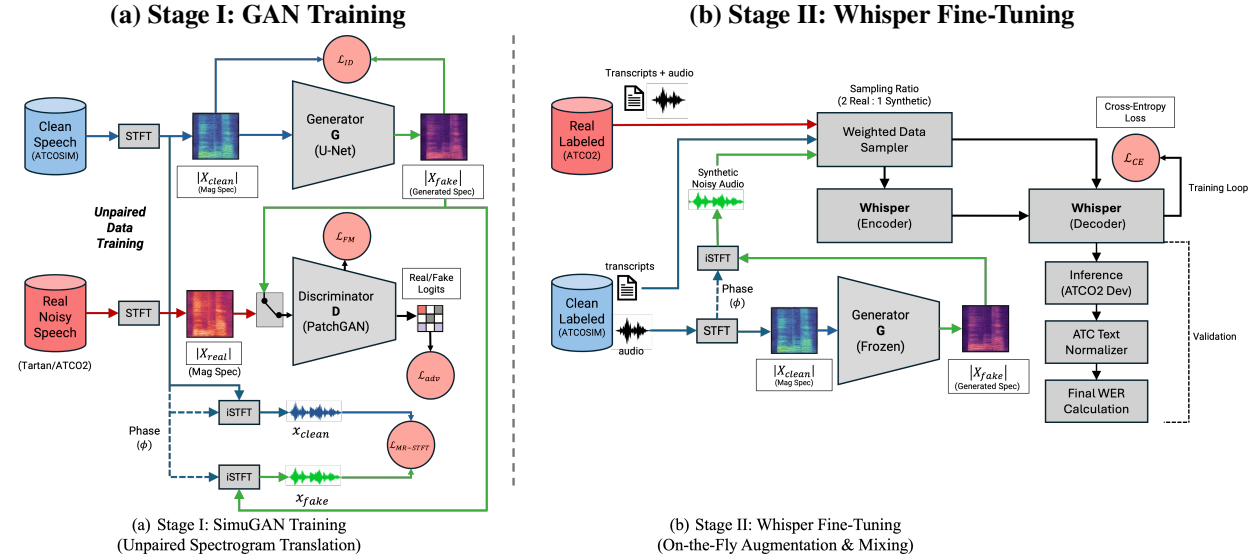


Fig. 1 The VHF-Sim2Real Framework. A vertical dashed line separates the two phases. (a) Left: We first train a Generative Adversarial Network (GAN) using unpaired data to map clean simulation audio (ATCOSIM) to the noisy VHF domain. Feature-matching ($\mathcal{L}_{FM}$), identity ($\mathcal{L}_{ID}$), and time-domain ($\mathcal{L}_{MR-STFT}$) losses encourage realistic noise generation while preserving speech content. (b) Right: We use the frozen Generator to augment clean data on-the-fly, mixing it with real labeled data (via a weighted data sampler with an approximate 2:1 real-to-synthetic sampling ratio) to robustly fine-tune Whisper. *Color legend:* Blue arrows denote clean simulation data, Red arrows denote real noisy data, and Green arrows denote synthetic noisy data. Dashed lines indicate phase information preserved for reconstruction.

A. Stage I: GAN Training

The objective of the first stage is to learn a mapping function $G : X_{\text{clean}} \rightarrow X_{\text{vhf}}$ that injects realistic VHF artifacts into clean spectrograms without requiring paired training data.

Input Representation Prior to entering the network, all audio is converted to the frequency domain via Short-Time Fourier Transform (STFT). The magnitude spectrograms ($|X_{\text{clean}}|$ and $|X_{\text{real}}|$) serve as the inputs to the GAN. The corresponding phase information (ϕ) is extracted and set aside (dashed blue path in Fig. 1a). The generator operates only on magnitudes to learn noise textures, and during waveform reconstruction we reuse the original phase via inverse STFT. This approximation ignores the phase characteristics of additive channel noise, but it keeps the synthesized waveforms time-aligned with the input utterances and avoids additional phase artifacts.

Network Architecture The generator G is a custom U-Net architecture designed for spectral translation. It features a three-stage encoder-decoder structure with a base channel width of 64, expanding to 256 in the bottleneck to capture global noise distributions. When operating on linear-magnitude spectrograms, we apply a Softplus activation in the final layer so that the predicted magnitudes remain non-negative. The discriminator D utilizes a PatchGAN architecture with spectral normalization applied to all convolutional layers to constrain the Lipschitz constant and stabilize adversarial training while focusing on local time-frequency structure.

Composite Loss Function To ensure the generated spectrograms capture the texture of VHF noise while preserving the linguistic content of the original speech, we optimize a composite objective function:

$$\mathcal{L}_G = \mathcal{L}_{\text{adv}}(G, D) + \lambda_{\text{id}} \mathcal{L}_{\text{id}} + \lambda_{\text{fm}} \mathcal{L}_{\text{fm}} + \lambda_{\text{mr-stft}} \mathcal{L}_{\text{mr-stft}}. \quad (1)$$

where,

- $\mathcal{L}_{\text{adv}}$ is the Least Squares GAN (LSGAN) loss, which provides smoother gradients for the generator than the original minimax formulation and drives the synthetic spectra toward the distribution of real VHF spectra.
- $\mathcal{L}_{\text{id}}$ is an identity or content-preservation loss that penalizes L_1 element-wise differences between the input clean spectrogram and the generated output. With a relatively small weight λ_{id} , it acts as a regularizer that discourages the generator from distorting phonetic content or severely altering the spectral envelope, while still allowing additive noise to be injected.
- $\mathcal{L}_{\text{fm}}$ is a feature-matching loss that aligns the intermediate feature representations of D for real and synthetic samples, encouraging the generator to match higher-order statistics of the VHF noise domain rather than only fooling D at its output layer.
- $\mathcal{L}_{\text{mr-stft}}$ is a multi-resolution STFT loss computed in the time domain. By reconstructing the waveform from the generated magnitudes and the original phase and comparing its multi-scale STFTs to a clean reference, this term encourages the synthetic audio to remain temporally coherent and to preserve the coarse structure of the speech signal while permitting local perturbations that correspond to channel noise.

In combination, the adversarial and feature-matching terms pull the generator toward the real VHF noise distribution, while the identity and multi-resolution STFT losses act as content-aware regularizers that keep the injected noise from overwhelming the underlying utterance.

B. Stage II: Whisper Fine-Tuning

In the second stage, we fine-tune the Whisper large-v2 model. Training solely on synthetic data risks a mismatch with real radio conditions, while training solely on the limited real data risks overfitting. To mitigate this, we employ two key strategies.

On-the-Fly Augmentation During fine-tuning, the generator G is frozen. For every batch, ATCOSIM utterances are first transformed to magnitude spectrograms and passed through G to synthesize noisy variants in real time. These synthetic spectra are then combined with the original phase and converted back to waveforms via inverse STFT. This on-the-fly generation prevents the ASR model from memorizing a fixed augmented dataset and exposes it to a wider variety of noise realizations over the course of training.

Weighted Data Sampling We implement a weighted data sampler to control the mixture of training examples. It constructs batches from two streams of audio–transcript pairs: (1) real ATCO2 utterances with their human transcripts and (2) ATCOSIM utterances, whose audio is optionally passed through the generator to obtain noisy variants while retaining the original simulation transcripts. The sampler assigns higher sampling probability to the ATCO2 stream so that, on average, roughly two thirds of the items in a batch come from real ATCO2 and one third from synthetic ATCOSIM. Because the labeled ATCO2 subset is much smaller than ATCOSIM, this strategy deliberately oversamples each ATCO2 utterance relative to each ATCOSIM utterance, anchoring the model to the target domain distribution at the cost of reusing the same real examples multiple times. In practice, we found that this approximate 2:1 real-to-synthetic sampling ratio provided a good balance between leveraging the acoustic diversity of GAN-augmented simulation data and maintaining performance on real VHF communications.

V. Implementation Details

We implemented the VHF-Sim2Real framework in PyTorch. All experiments were conducted on a single NVIDIA L40 GPU with 48 GB of VRAM on the Georgia Tech PACE Phoenix cluster. Table 2 summarizes the hyperparameters and training configurations used in our experiments.

A. GAN Training Configuration

To stabilize adversarial training, we used the optimization setup summarized in the Phase I section of Table 2. In particular, we applied a two-time-scale update rule (TTUR) with a higher learning rate for the discriminator than for the generator, so that the discriminator can track changes in the generator and maintain a useful training signal. We also used one-sided label smoothing with $y_{\text{real}} = 0.9$ to reduce overconfident discriminator predictions early in training and mitigate vanishing gradients for the generator.

Table 2 Complete hyperparameter configuration for the VHF-Sim2Real framework. The upper section details the adversarial training parameters, while the lower section outlines the ASR fine-tuning specifications.

Parameter	Value
<i>Phase I: GAN Training</i>	
Optimizer	Adam ($\beta_1 = 0.5, \beta_2 = 0.999$)
Learning Rate	$G : 1 \times 10^{-4}, D : 2 \times 10^{-4}$ (TTUR)
Batch Size	8
Epochs	15 (Step Decay: $\gamma = 0.5$ every 5 epochs)
Loss Weights	$\lambda_{\text{id}} = 4, \lambda_{\text{fm}} = 10, \lambda_{\text{mr-stft}} = 1$
Label Smoothing	$y_{\text{real}} = 0.9$ (One-sided)
<i>Phase II: Whisper Fine-Tuning</i>	
Base Model	Whisper large-v2 (1.55B params)
Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$)
Learning Rate	1×10^{-5} (Cosine decay with 500 warmup steps)
Batch Size	8 (effective)
Epochs	10
Precision	Mixed precision (FP16)
Data Mix	Approx. 2:1 (Real : Synthetic, via probabilistic sampling)

B. Whisper Fine-Tuning Configuration

We initialized our ASR model with the pretrained Whisper large-v2 checkpoint and fine-tuned it using the Hugging Face Transformers library with the specifications listed in the Phase II section of Table 2. To adapt the model conservatively and avoid large deviations from the pre-trained parameters, we used a small learning rate with a cosine decay schedule. Gradient checkpointing and mixed precision (FP16) training were enabled to reduce memory footprint and improve throughput, which allowed us to maintain a reasonable effective batch size and context length on a single L40 GPU.

C. Evaluation Pipeline

During inference, we used greedy decoding. To compute the ATC-specific WER, we applied a text normalization pipeline adapted from [9]. This process handles:

- 1) **Case and token normalization:** Transcripts are lowercased and common orthographic variants are unified, for example “X-ray” and “xray” are both mapped to “xray”.
- 2) **Number normalization:** Spelled-out and numeric expressions are converted to space-separated digit tokens, for example “ninety five” or “95” are mapped to “9 5” and “one zero two two” or “1022” are mapped to “1 0 2 2”.
- 3) **Callsign normalization:** Airline identifiers are standardized to their spoken callsigns (for example “DAL” is mapped to “delta”) so that scoring focuses on semantic mistakes rather than formatting differences.

Following normalization, the word error rate (WER) is calculated as

$$\text{WER} = \frac{S + D + I}{N} \quad (2)$$

where S , D , and I denote substitutions, deletions, and insertions, and N is the total number of reference tokens. Table 3 illustrates the impact of this normalization on a sample transcript.

Table 3 Example of ATC-specific text normalization applied to an evaluation transcript.

Stage	Transcript Example
Reference	<i>Hotel November X-ray ninety five crossing is approved QNH one zero two two</i>
Normalized	<i>hotel november xray 9 5 crossing is approved qnh 1 0 2 2</i>

VI. Evaluation and Results

We evaluate the VHF-Sim2Real framework in two ways: (1) by analyzing the acoustic properties of the synthesized noise to check that it resembles key aspects of real VHF channels, and (2) by measuring the downstream impact on ASR accuracy on the ATCO2 benchmark.

A. Acoustic Fidelity of GAN-Generated Noise

To assess whether the generator learns realistic VHF artifacts, we examine the spectral characteristics of its output. Fig. 2 presents a side-by-side comparison of clean, GAN-synthesized, and real noisy spectrograms.

Spectral Analysis Two main effects are apparent in the generated spectrograms. First, the GAN introduces narrowband, transient interference patterns that appear as vertical streaks and resemble squelch bursts and transmitter keying artifacts. Second, the harmonic structure of the speech remains clearly visible, which is consistent with the content-preserving losses constraining the generator not to overwrite linguistic information. Compared with the clean simulation, energy is more concentrated in the lower part of the spectrum and reduced at higher frequencies, although the synthetic spectrogram still retains more high-frequency energy and a lower stationary noise floor than the real ATCO2 target. This suggests that the model functions primarily as an additive noise simulator that injects VHF-like transients and partial band limiting rather than reproducing the full combination of lowpass filtering and continuous background static seen in operational recordings.

Objective Metrics We evaluate the generator on 500 held-out ATCOSIM utterances and their GAN-augmented counterparts, together with a matched sample of real ATCO2 audio (Table 4). Metrics that require paired signals, including Log-Spectral Distance (LSD), multi-resolution STFT loss, SI-SDR, STOI, embedding distance, speaker distance, and SNR_{KL} , are computed between each clean ATCOSIM utterance and its corresponding synthetic version. Fréchet Audio Distance (FAD) is computed distributionally between the set of synthetic utterances and real ATCO2 recordings, following Kilgour et al. [47]. In this way, the paired metrics quantify how much linguistic and speaker information is preserved when noise is added, while FAD reflects how close the overall synthetic distribution is to the real VHF domain.

The resulting values support two main design goals:

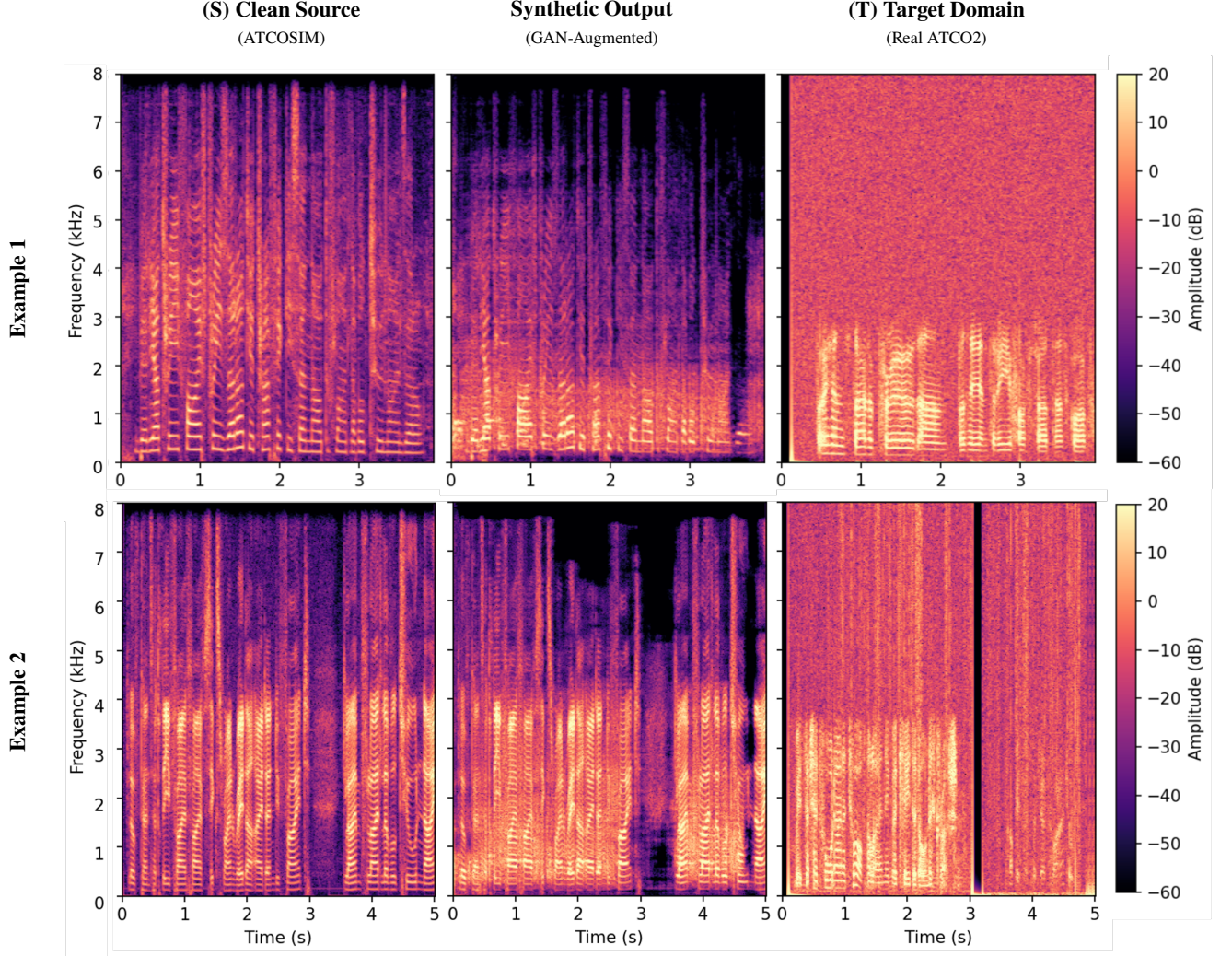


Fig. 2 Spectrogram comparison demonstrating domain translation. The GAN output (center) introduces characteristic VHF-like artifacts, in particular transient vertical interference events and partial high-frequency attenuation, into the clean simulation (left). The synthetic spectra share several qualitative patterns with the real ATCO2 target (right), such as intermittent bursts and reduced high-frequency speech content, but exhibit a lower stationary noise floor and somewhat more high-frequency energy.

- 1) **Content preservation:** The low embedding distance (0.11), speaker distance (0.113), and STOI score of 0.83, together with an SI-SDR of 3.96 dB, indicate that the GAN modifies the signal enough to introduce audible channel effects while largely retaining phonetic content and speaker identity.
- 2) **Plausible noise injection and distributional shift:** The LSD of 10.43 dB, multi-resolution STFT loss of 0.068, SNR_{KL} of 15.31, and a FAD of 35.73 show that the augmented signals differ substantially from the clean inputs, while the FAD of 35.73 indicates that, as a distribution, the synthetic audio moves closer to the statistics of real VHF recordings rather than leaving the spectra essentially unchanged.

B. ASR Performance (ATCO2 Benchmark)

We evaluate recognition accuracy on the held-out ATCO2 test set using the ATC-specific text normalization described in Section V. For our baseline reproduction and VHF-Sim2Real system, all WER values are computed with the same decoding and normalization pipeline. For reference, we also include the zero-shot performance of Whisper large-v2 (29.05% WER) as reported by van Doorn et al. [9].

Table 4 Objective evaluation of GAN-synthesized audio quality (N=500). Arrows indicate the direction of better performance when the metrics are interpreted as indicators of speech preservation quality (for example higher SI-SDR and STOI denote more intelligible speech, lower LSD and FAD denote closer match to the respective reference distributions).

Metric	Mean	CI (95%)	Metric	Mean	CI (95%)
LSD (↓)	10.43	±0.09	EmbedDist (↓)	0.11	±0.01
MR-STFT (↓)	0.068	±0.002	SpkDist (↓)	0.113	±0.006
SI-SDR (↑)	3.96	±0.25	SNR _{KL} (↓)	15.31	±0.11
STOI (↑)	0.83	±0.004	FAD (↓)	35.73	–

To establish a baseline, we replicated the fine-tuning strategy of [9] using a mix of clean simulation and real training data. This baseline configuration yields a WER of 14.32%, which is consistent with the findings reported by van Doorn et al. and indicates that standard fine-tuning only partially closes the gap between simulated and real VHF speech. When GAN-augmented ATCOSIM data are added to the training loop, the WER on the same test set decreases to 3.58%. This corresponds to a relative error reduction of 75.0%. In our setting, this improvement indicates that acoustic mismatch is a major contributor to the sim-to-real performance gap and that unpaired generative augmentation can substantially reduce this gap on the ATCO2 benchmark. More broadly, the results suggest that learning VHF noise transformations from unpaired radio recordings is a promising approach for aviation ASR when labeled operational data are scarce, although further evaluation on independent corpora will be needed to characterize how well these gains generalize beyond the one-hour public ATCO2 subset.

Table 5 WER comparison on the real-world ATCO2 test set. The zero-shot Whisper result is taken from van Doorn et al. [9]; the remaining rows are from our runs on the ATCO2-test-set-1h evaluation split.

Model / Approach	WER (%)	Rel. Improv.
Zero-shot Whisper (Large-v2, van Doorn et al.)	29.05	–
Baseline (Clean + Real), our reproduction	14.32	–
VHF-Sim2Real (ours)	3.58	+75.0%

VII. Limitations

Our study is subject to several limitations in terms of data coverage and modeling assumptions. From a data perspective, all experiments rely on three open ATC corpora (ATCOSIM, TartanAviation, and the public ATCO2 subset) that cover only a narrow subset of real-world airports and airspaces. Noise characteristics, radio equipment, and local phraseology vary across regions and sectors, so the gains we observe on this combination of European simulation data and predominantly US terminal-area noise may not transfer directly to other operating environments. The GAN is trained to approximate an average VHF channel based on these datasets rather than airport- or channel-specific conditions, and practical deployments would likely require additional adaptation to local microphones, radios, and traffic patterns as more labeled data become available.

From a signal processing standpoint, our generation pipeline operates on magnitude spectrograms and reconstructs waveforms using the clean source phase. This approximation ignores channel effects that alter the phase of the signal, such as non-linear clipping, fading, and multi-path interference, and can produce synthetic audio that is somewhat more phase coherent than genuine high-interference transmissions. In addition, visual inspection of the spectra suggests that the GAN prioritizes transient interference patterns, for example squelch bursts and short keying artifacts, while underestimating the continuous stationary noise floor and strict lowpass characteristics of analog VHF receivers. As a result, the augmented data may capture salient transients that are useful for training but still underrepresent some of the most challenging background conditions found on operational channels.

There are also limitations associated with the training and evaluation protocol. The labeled ATCO2 subset available for supervised fine-tuning is only about one hour, so our weighted sampling strategy necessarily oversamples these utterances to anchor the model in the target domain. This heavy reuse of a small set of speakers and channel conditions increases the risk of overfitting to that particular test environment, even though all baselines and ablations are evaluated

under the same scoring pipeline. Finally, all reported results are obtained on a single held-out ATCO2 partition and a single ASR backbone, Whisper large-v2. Further experiments on independent ATC corpora and alternative recognizers will be needed to determine how well the observed improvements generalize beyond this specific setting.

VIII. Future Work

These limitations suggest several directions for future work. On the modeling side, a natural next step is to move beyond magnitude-only processing with clean phase reconstruction. Time-domain generative models or complex-valued spectrogram architectures would allow the generator to represent phase-destructive channel effects, including clipping, fading, and multi-path interference, that our current magnitude-only formulation can only approximate. Additionally, explicitly incorporating standard analog receiver responses into the training objective or conditioning could encourage the generator to respect the strict bandwidth limits of VHF communications rather than leaving high-frequency content unconstrained.

A more detailed analysis of which artifacts the GAN reproduces faithfully, and which remain difficult to model, would also support more principled use of the available data. For example, automatic scoring of synthetic samples by artifact type and severity could be used to drive curriculum-style training schedules, where the ASR model is first exposed to simpler noise patterns and then progressively to more challenging channel conditions as its robustness improves.

Beyond acoustics, combining VHF-Sim2Real with downstream modules that operate on transcripts is a promising direction. For instance, natural language understanding components or domain-specific rule engines could use improved transcripts to flag inconsistent read-backs or missing clearances. Quantifying how residual recognition errors propagate into such higher-level safety logic, and how much ASR robustness is sufficient for reliable monitoring, remains an open question.

IX. Conclusion

We presented *VHF-Sim2Real*, a generative augmentation framework that uses unlabeled VHF radio recordings to expand the amount of labeled ATC speech that exhibits realistic channel degradations. By training a spectrogram-based GAN to map clean ATCOSIM utterances into a VHF-like domain and then fine-tuning Whisper on a mixture of real ATCO2 speech and GAN-augmented simulation, we reduce the WER on the public ATCO2 test set from 14.32% for a standard fine-tuning baseline to 3.58%, a 75.0% relative reduction obtained with the same labeled ATC data.

Our analyses indicate that the generator reproduces characteristic transient interference patterns of VHF channels while largely preserving speaker and phonetic content, although it underestimates some aspects of the stationary noise floor and bandwidth limitations. Together with the oversampling of a small labeled target subset, these modeling choices mean that the reported gains should be interpreted as evidence of what is achievable in this specific setting rather than as a definitive bound on ATC ASR performance. Nevertheless, the results demonstrate that learning channel effects from large unlabeled radio corpora and using them for unpaired generative augmentation is a practical and effective way to strengthen aviation ASR systems without requiring a proportional increase in manual transcription effort.

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